

Electronic Annex B – Supplementary material, guide to analytical databases and reference list

**Accompanying the article: Gallium, germanium, indium, and other trace
and minor elements in sphalerite as a function of deposit type – A
meta-analysis**

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1. Supplementary material

Table B1 – Loadings of all principal components (complementary to Fig. 6)

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
LN(Ag)	-0.098	-0.438	-0.139	-0.481	-0.676	-0.093	0.204	-0.180	-0.095
LN(Cd)	0.022	-0.204	0.638	-0.488	0.467	-0.142	0.260	-0.076	-0.024
LN(Co)	-0.134	-0.178	-0.662	-0.315	0.537	-0.030	-0.155	-0.180	-0.259
LN(Cu)	-0.057	-0.620	0.094	-0.003	0.051	0.350	-0.470	0.434	0.263
LN(Fe)	-0.490	-0.080	-0.122	0.116	0.068	-0.657	0.120	0.210	0.480
LN(Ga)	0.403	-0.334	0.098	0.299	0.008	-0.414	-0.370	-0.559	0.079
LN(Ge)	0.496	-0.170	-0.114	0.084	0.007	-0.388	0.175	0.597	-0.409
LN(In)	-0.262	-0.446	0.016	0.562	0.125	0.224	0.516	-0.159	-0.244
LN(Mn)	-0.502	0.070	0.298	0.074	-0.117	-0.211	-0.445	0.050	-0.625

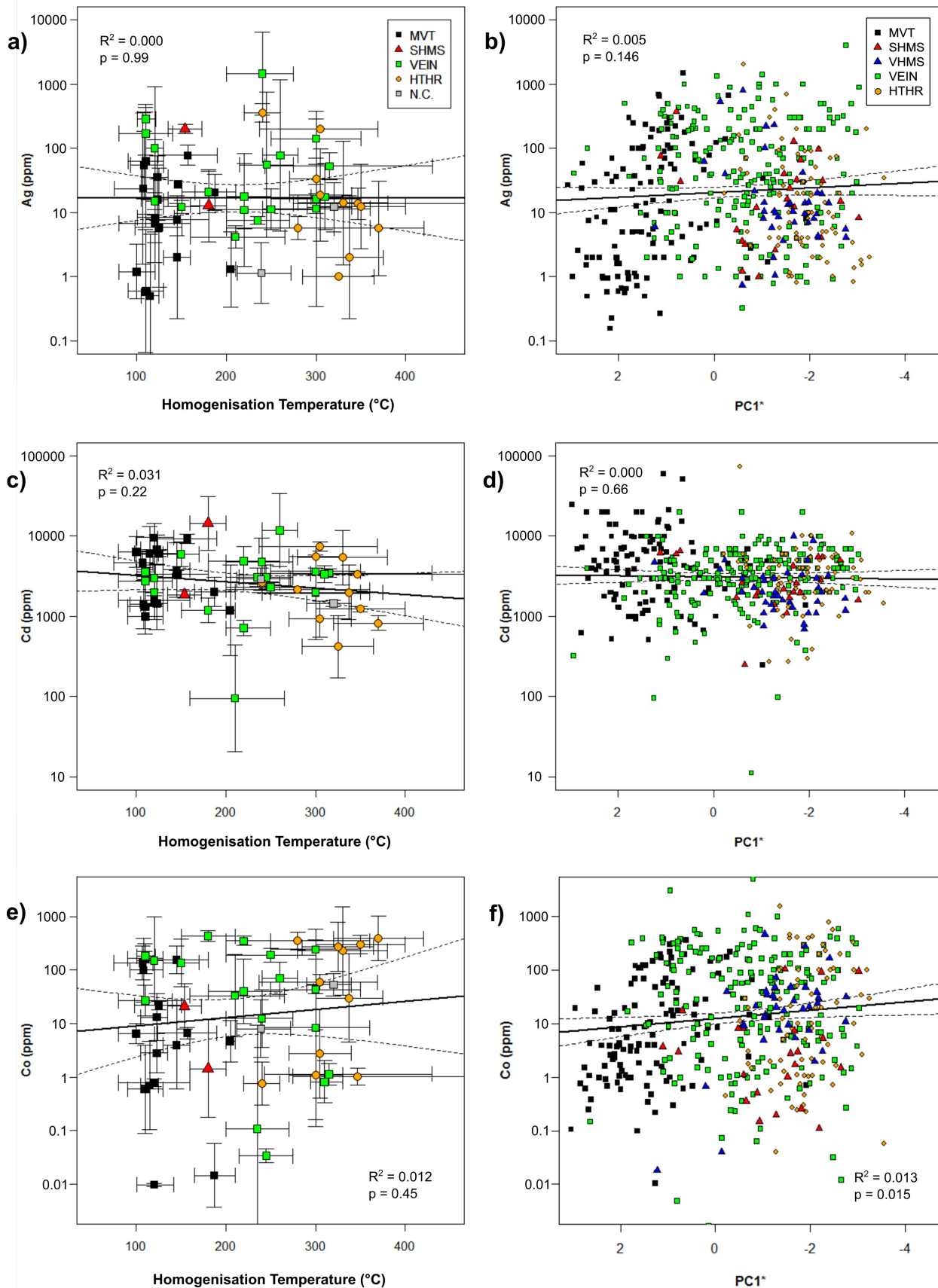


Fig. B1: Relationship of trace element concentrations in sphalerite with temperature and PC1*: a and b) Ag, c and d) Cd, e and f) Co, and overleaf, g and h) Cu, i and j) Fe, and k and l) Mn. Correlation coefficients and p-values shown correspond to linear relationships of temperature and PC1* with the log-transformed element concentrations. Lines of best fit are shown as bold black lines, while 95% confidence intervals are shown as broken lines. Note that the plots of

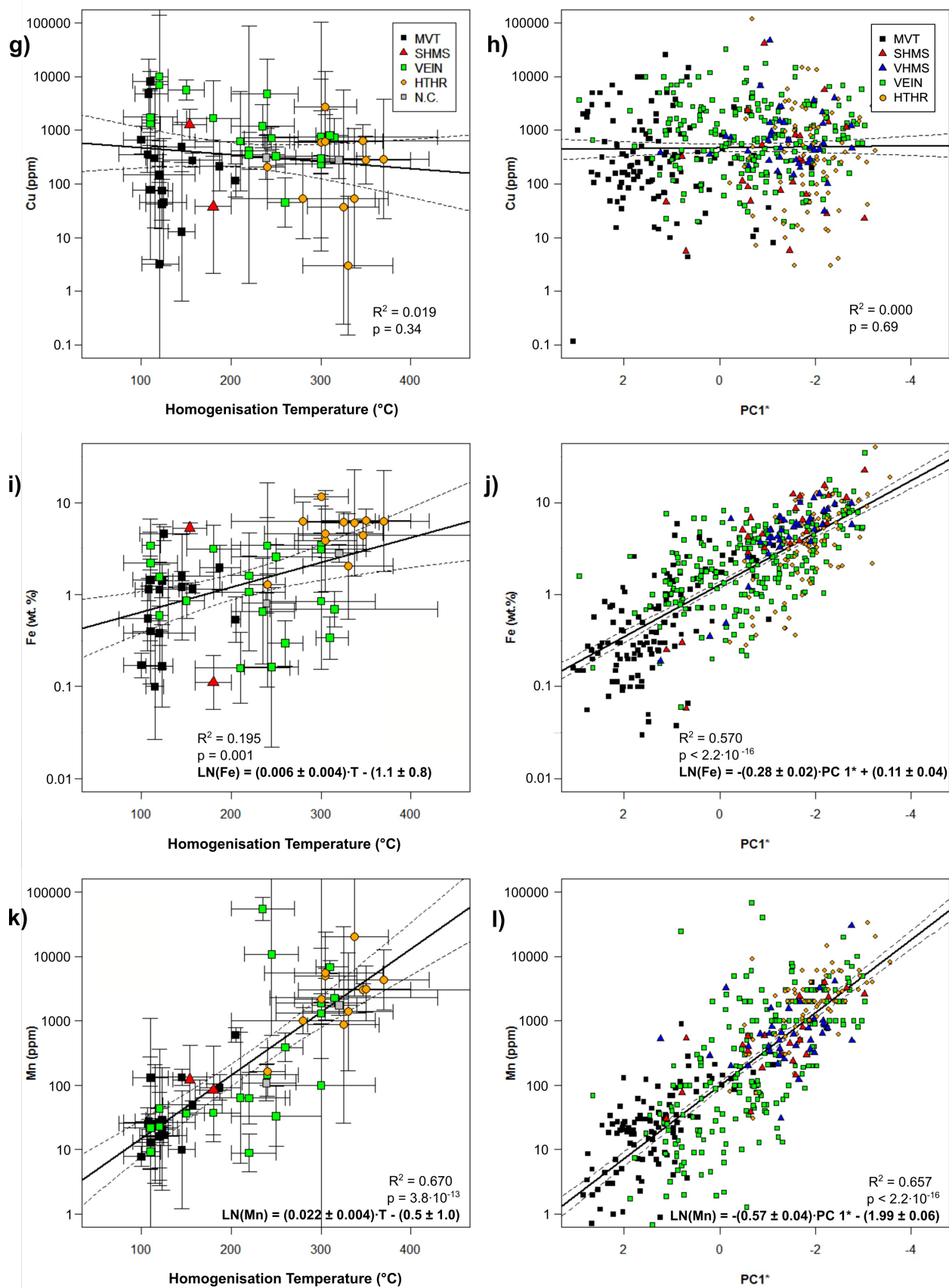


Fig. B1 (cont'd): element concentrations vs. PC1* show the whole dataset (including imputed values), while the plots of element concentrations vs. temperature only show those datapoints for which thermometric information was available. The range in PC1* values shown on the y-axes of panels b, d, f, h, j and l corresponds exactly to the range in temperatures shown in panels a, c, e, g, i and k (conversion via the fitted relationship shown in Fig. 11a).

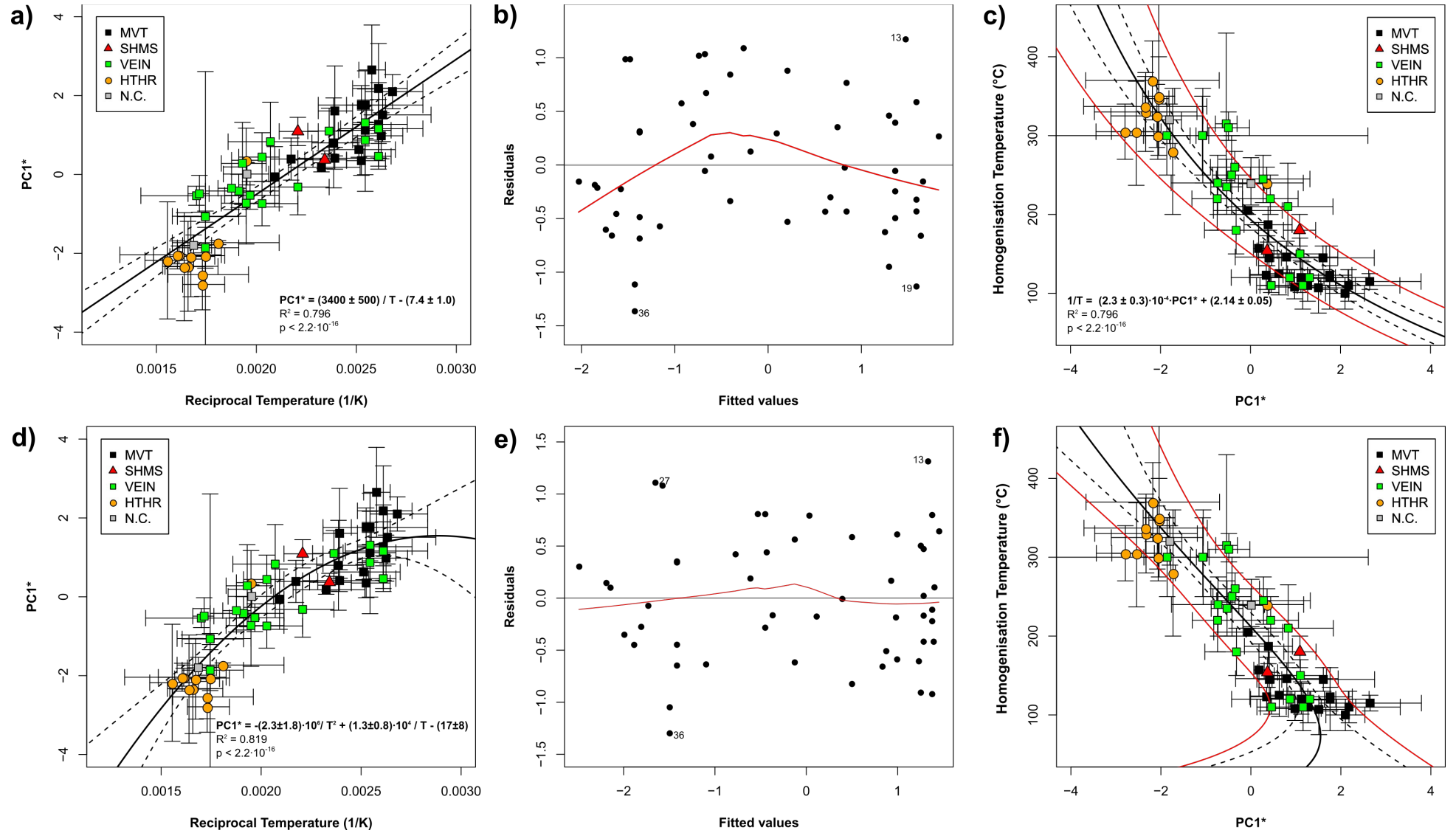


Fig. B2: Different fits to the relationship between PC1* and temperature, assuming a) to c) a linear relationship between PC1* and 1/T; and d) to f) a quadratic relationship between PC1* and 1/T. A possible thermodynamic justification for these two types of fit is given in detail in the main text. b) and e) show the residual plots for the fits provided in a) and d), while c) and f) show the corresponding prediction intervals for a tentative geothermometer analogous to the one illustrated in Fig. 11a.

2. Guide to analytical databases

The main purpose of this document is to provide a short introduction to the structure of the sphalerite database used for the article referred to above. It furthermore contains a detailed list of all references cited in the database. The database itself is available in .xls format as Electronic Annex A.

Database structure

Four spreadsheets are included in Electronic Annex A, the first carrying only publication information, the second carrying all unprocessed data points for analytical data, the third carrying the detailed thermometric data summarised in Table 3 and the fourth carrying the metamorphic grade and chemical data used in the compilation of Fig. 10. In the following, we give a short description of the fields used to characterise data points in the last three spreadsheets:

Analytical_Data spreadsheet

X (ppm) – the concentration of element X in sphalerite in units of ppm as reported in the literature.

DL – X (ppm) – the detection limit for element X, as cited in the publication from which the data was taken. If no detection limit was cited, a likely value is indicated which was inferred from both the way in which data is presented in the publication as well as a general knowledge of the analytical technique used. A question mark behind the value indicates where such a procedure was necessary.

Analytical technique – as cited in the publication from which the data was taken. Several techniques are listed if the data points used for calculating the average for a deposit were taken from different publications.

Appr. Error - the relative error of measurement as cited in the publication from which the data was taken (given as a fraction of 1). If no error was cited, a likely value is indicated which was inferred from both the way in which data is presented in the publication as well as a general knowledge of the analytical technique used. A question mark behind the value indicates where this was necessary.

No. of samples – the number of samples analysed to yield the value cited under X (ppm).

Locality – the locality where the analysed sample(s) originated, as cited in the original publication.

District/Region/Formation – the mining district, region or geological formation in which the locality from which the sample(s) was (were) taken is located. Usually given as cited in the original publication.

State/Country – usually given as cited in the publication from which the data was taken. Otherwise inferred from the literature search.

Continent – the continent on which the deposit the sample(s) was (were) taken from is located.

Geology – the type of deposit, determined according to the classification set out in the main text.

Reference (Chemistry) – the publication(s) from which the chemical data was taken.

Reference (Geology) – the publication(s) from which the geological data was taken.

Notes – comments on values entered in earlier columns of the data table.

Error – X - the relative error of measurement for element X as cited in the publication from which the data was taken (given as a fraction of 1).

Thermometric_Salinity_Data spreadsheet

Locality – the locality from which the analysed samples originated (same as on the Analytical_Data spreadsheet).

District – the mining district from which the analysed samples originated (same as on the Analytical_Data spreadsheet).

State/Country - the state/country from which the analysed samples originated (same as on the Analytical_Data spreadsheet).

TYPE – the type of deposit (same as on the Analytical_Data spreadsheet).

N – No. of samples analysed chemically.

PC_i – the mean value of the *i*th principal component (PC), as determined from the data in the Analytical_Data spreadsheet.

PC_i_error – the estimated error on the mean value of *PC_i*. This includes measurement, sampling as well as imputation errors where applicable (see main text). The value given corresponds to twice the standard error, i.e. it defines a 95 % confidence interval.

Ga_Ge – the mean of the log-ratio $\ln(\text{Ga/Ge})$ for all samples.

Ga_Ge_error – the error on *Ga_Ge*, again including measurement, sampling and imputation errors where applicable (cf. main text). The value given corresponds to twice the standard error, i.e. it defines a 95 % confidence interval.

T – the best estimate of the median temperature of formation or, in most cases, the median homogenisation temperature of fluid inclusions directly associated to ore mineralisation. Therefore, most of the values cited in this column are minimum formation temperatures.

T_U – the upper end of the confidence interval for *T*. Usually, the confidence interval was taken as $T \pm$ one standard deviation of the values measured on single fluid inclusions, or alternatively as the interval cited in the respective publication. This is thought to give a good approximation to the 95 % confidence interval. For a more detailed discussion see main text.

T_L – the lower end of the confidence interval for *T*.

S – the best estimate of the mean salinity of the fluid involved in the formation of the deposit, in most cases the salinity of fluid inclusions directly associated to ore mineralisation.

S_U – the upper end of the confidence interval for *S*. The confidence interval for *S* was defined in a similar manner to that for *T* above.

S_L – the lower end of the confidence interval for *S*.

Method – the method by which *T* and *S* were determined. See Table 3 for abbreviations.

Mineral(s) – the mineral(s) used for the determination of S and T.

REF_CHEM – the reference(s) from which the analytical data for the estimates of the PCi was taken.

REF_TEMP – the reference(s) from which the data for T and S was derived.

Elements not analysed – elements not analysed for the datapoint in question. The values for these elements were imputed according to the procedures described in detail in the main text.

Metamorphic_Grade_Data spreadsheet

Locality – as before, the locality from which the samples were taken.

State/Country – the state/country in which the locality is located.

TYPE – the type of deposit (same as on the Analytical_Data spreadsheet).

Facies – the metamorphic facies of the deposit.

N – No. of samples analysed chemically.

PC1 – the mean value of the first principal component (PC), as determined from the data in the Analytical_Data spreadsheet.

PC1_error – the estimated error on the mean value of PCi. This includes measurement, sampling and imputation errors where applicable (see main text). The value given corresponds to twice the standard error, i.e. it defines a 95 % confidence interval.

T_calc – the (minimum) temperature of formation/equilibration calculated from the GGIMFIS geothermometer using the calibration shown in Fig. 11.

T_calc_U – the upper end of the confidence interval for T_calc. This was estimated using both the confidence interval for PC1, as well as the estimation interval shown on Fig. 11.

T_calc_L – the lower end of the confidence interval for T_calc.

T_peak – best estimate of the peak metamorphic temperature, i.e. the highest temperature the deposit was heated to during metamorphism.

T_peak_U – the upper end of the confidence interval for T_peak.

T_peak_L – the lower end of the confidence interval for T_peak.

REF_CHEM – the reference(s) from which the analytical data for the estimates of the PCi was taken.

REF_MET – the reference(s) from which the data for T_peak was derived.

Elements not analysed – elements which were not analysed in REF_CHEM and therefore had to be imputed for the estimation of PC1.

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